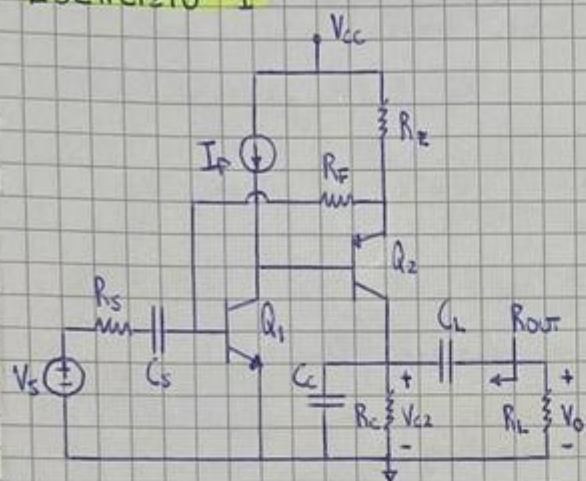


ESERCIZIO 1



Consider the two-stages amplifier in Fig. 1. The circuit parameters, at $T = 25^\circ$, are:

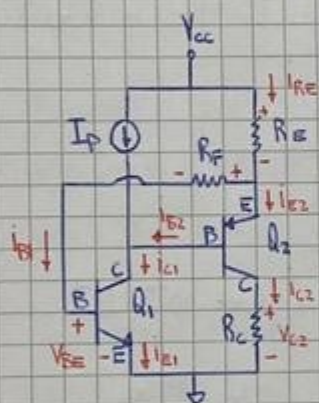
$$V_{CC} = 24 \text{ V} \quad I_P = 1 \text{ mA} \quad R_S = 47 \text{ k}\Omega \\ R_C = 12 \text{ k}\Omega \quad R_L = 10 \text{ k}\Omega \quad R_F = 1,5 \text{ M}\Omega$$

$$Q_1: V_{BE} = 0,7 \text{ V} \quad \beta_F = \beta_0 = 100 \quad r_o = +\infty$$

$$Q_2: V_{BE} = -0,7 \text{ V} \quad \beta_F = \beta_0 = 100 \quad r_o = +\infty$$

$$V_T = 25 \text{ mV}$$

- 1) Dc analysis: • compute R_E value knowing that $V_{C2} = 11 \text{ V}$
• compute Q_1 and Q_2 operating points (V_{CE}, I_C).



$$V_{C2} = R_C I_{C2} = 11 \text{ V} \Rightarrow I_{C2} = V_{C2} / R_C = 0,92 \text{ mA}$$

$$I_{B2} = I_{C2} / \beta_F = 9,2 \text{ }\mu\text{A} \quad I_{E2} = (\beta_F + 1) I_{B2} = 0,93 \text{ mA}$$

$$I_{C1} = I_{B2} + I_P = 1,01 \text{ mA} \quad I_{B1} = I_{C1} / \beta_F = 10,1 \text{ }\mu\text{A}$$

$$I_{E1} = (\beta_F + 1) I_{B1} = 1,02 \text{ mA} \quad I_{RE} = I_{E2} + I_{B1} = 0,94 \text{ mA}$$

$$V_{CC} - I_{RE} R_E - I_{B1} R_F - V_{BE} = 0$$

$$\Rightarrow R_E = \frac{V_{CC} - V_{BE} - I_{B1} R_F}{I_{RE}} = 8,67 \text{ k}\Omega$$

$$V_{CE1} = V_{CC} - R_E I_{RE} + V_{BE} = 15,15 \text{ V} > V_{CESAT} \rightarrow \text{OK } Q_1 \text{ in zsd}$$

$$V_{CE2} = (V_{CC} - R_E I_{RE}) - R_C I_{C2} = 4,81 \text{ V} > V_{CESAT} \rightarrow \text{OK } Q_2 \text{ in zsd}$$

$$\text{OP } Q_1 (15,15 \text{ V}, 1,01 \text{ mA}) \quad \text{OP } Q_2 (4,81 \text{ V}, 0,92 \text{ mA})$$

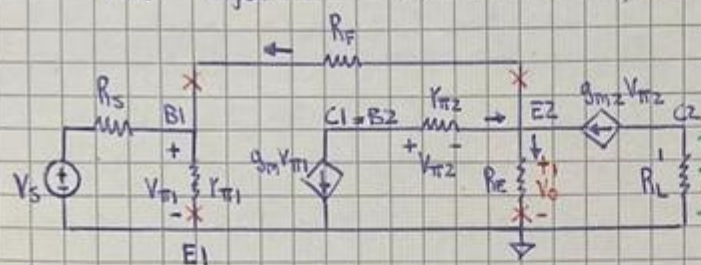
Parametri di piccoli segnali:

$$g_{m1} = 40 I_{C1} = 40,4 \text{ mS} \quad r_{\pi 1} = \beta_0 / g_{m1} = 2,5 \text{ k}\Omega \quad r_{o1} \rightarrow +\infty$$

$$g_{m2} = 40 I_{C2} = 36,8 \text{ mS} \quad r_{\pi 2} = \beta_0 / g_{m2} = 2,7 \text{ k}\Omega \quad r_{o2} \rightarrow +\infty$$

NB: C_S, C_L condensatori di accoppiamento (capacità grande)
 C_C condensatore di bypass alle alte frequenze (capacità piccola)

2) AC analysis: assuming C_s and C_L to be equivalent to short circuits and C_c to be equivalent to an open circuit at the frequency of interest. Determine the voltage gain $A_v = V_o/V_s$ of the amplifier and the output resistance R_{out} indicated in the figure. (simile a 10,72 Sedra Smith)



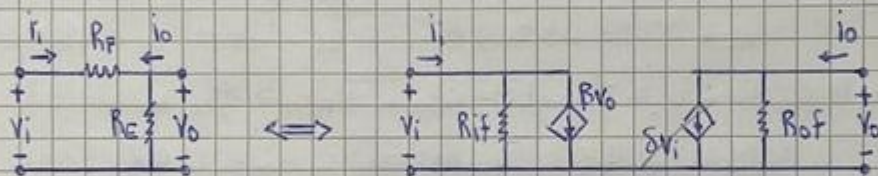
$$R_L' = R_c \parallel R_L = 5454,55 \, \Omega$$

Posso risolverlo in due modi, in base a dove considero l'uscita:

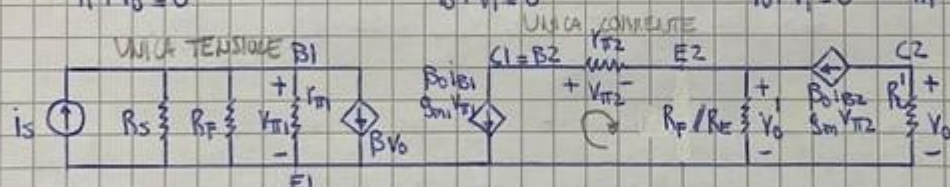
- 1) Dove è connesso il feedback (sempre voltage sampling!)
- 2) Segnata sul circuito (sempre current sampling!)

SOLUZIONE 1: feedback topology is trans-resistance because

- current/shunt mixing: $i_{B1} = i_s + i_{RF}$ which drives Q_1
- voltage/shunt sampling: V_o is the sampled variable. If R_E is shorted then $i_{RF} = 0$; instead if R_E is opened $i_{RF} \neq 0$



$$R_{if} = \frac{V_i}{i_i} \Big|_{V_o=0} = R_F \quad R_{of} = \frac{V_o}{i_o} \Big|_{V_i=0} = R_F \parallel R_E \quad \beta = \frac{i_i}{V_o} \Big|_{V_i=0} = -\frac{1}{R_F} = -0,67 \cdot 10^{-6} \, \Omega^{-1}$$



$$i_s = V_s/R_s$$

In open loop ($\beta \rightarrow 0$, $\beta = 0$):

$$V_{\pi 1} = (R_s \parallel R_F \parallel R_{E1}) i_s$$

$$V_o = (R_E \parallel R_F) (\beta_0 + 1) (-g_m V_{\pi 1})$$

$$\Rightarrow A_{OL} = \frac{V_o'}{i_s} \Big|_{\beta=0} = -g_m (\beta_0 + 1) (R_E \parallel R_F) (R_s \parallel R_F \parallel R_{E1}) = -83,36 \cdot 10^6 \, \Omega$$

In closed loop conditions:

$$A_F = \frac{V_o'}{i_s} = \frac{A_{OL}}{1 + \beta A_{OL}} = -1,47 \cdot 10^6 \, \Omega \approx \frac{1}{\beta} = -1,45 \cdot 10^6 \, \Omega \text{ due to } T = \beta A_{OL} = 55,9 \text{ high}$$

$$A_v = \frac{V_o}{V_s} = \frac{V_o}{V_o'} \cdot \frac{V_o'}{i_s} \cdot \frac{i_s}{V_s} = K_o A_F K_s = +13,6 \quad \checkmark$$

$$K_s = \frac{1}{R_s} = 2,1277 \cdot 10^{-5} \, \Omega^{-1}$$

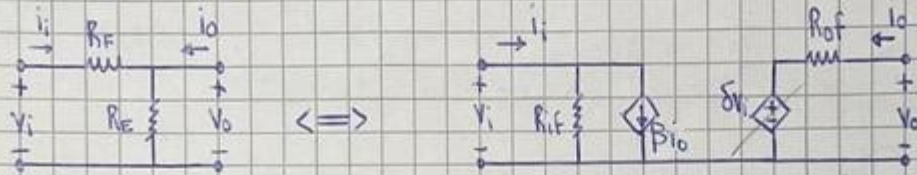
$$\text{To find } K_o: V_o' = (\beta_0 + 1) i_{B2} R_F \parallel R_E$$

$$V_o = -\beta_0 i_{B2} R_L'$$

$$\Rightarrow \frac{V_o}{V_o'} = \frac{-\beta_0 R_L'}{(\beta_0 + 1) R_F \parallel R_E} = -0,6266$$

SOLUZIONE 2: feedback topology is current amplifier because:

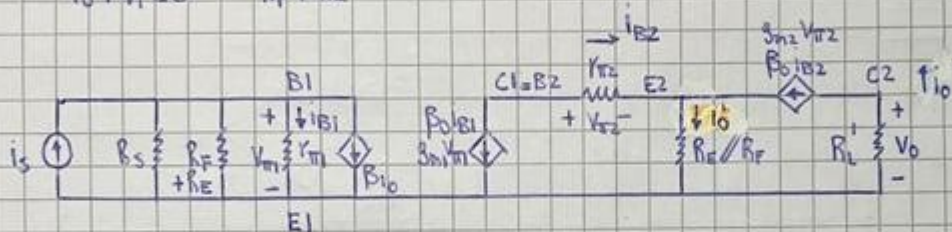
- current/shunt mixing: $I_{B1} = I_S + I_{RF}$ which drives Q_1
- current/series sampling: i_o is the sampled variable. If R_L' is opened then $i_o = 0$; instead if R_L' is shorted then $i_o \neq 0$.



$$R_{if} \triangleq \frac{V_i}{i_i} \Big|_{i_o=0} = R_F + R_E = 1,503 \text{ k}\Omega$$

$$R_{of} \triangleq \frac{V_o}{i_o} \Big|_{V_i=0} = R_E \parallel R_F = 8620 \text{ }\Omega$$

$$\beta \triangleq \frac{i_i}{i_o} \Big|_{V_i=0} = -\frac{R_E}{R_F + R_E} = -5,75 \cdot 10^{-3} \text{ A/A}$$



$$I_S = V_S / R_S$$

NB: i_o è quella che scorre su R_{of} !

In open loop conditions ($\delta \rightarrow 0, \beta = 0$):

$$V_{T1} = (R_S \parallel (R_E + R_F) \parallel r_{T1}) I_S \quad i_o' = (\beta_0 + 1) i_{B2} = -(\beta_0 + 1) g_{m1} V_{T1}$$

$$\Rightarrow A_{OL} \triangleq \frac{i_o'}{I_S} \Big|_{\beta=0} = -(\beta_0 + 1) g_{m1} (R_S \parallel (R_E + R_F) \parallel r_{T1}) = -9671 \text{ A/A}$$

In closed loop conditions:

$$A_F = \frac{i_o}{I_S} = \frac{A_{OL}}{1 + \beta A_{OL}} = -170,8 \approx \frac{1}{\beta} = -173,9 \text{ due to high } T = \beta A_{OL} = 55,6$$

$$A_V = \frac{V_o}{V_S} = \frac{V_o}{i_o} \cdot \frac{i_o}{I_S} \cdot \frac{I_S}{V_S} = K_0 A_F K_S = +19,6 \quad \checkmark$$

$$K_S = \frac{1}{R_S} = 2,1277 \cdot 10^{-5} \text{ }\Omega^{-1}$$

$$\text{To find } K_0: i_o = i_o' \frac{\beta_0}{\beta_0 + 1} \quad V_o = -R_L' i_o$$

$$\Rightarrow K_0 = \frac{V_o}{i_o} = -\frac{\beta_0}{\beta_0 + 1} R_L' = -5400,55 \text{ }\Omega$$

$$R_{out}^{OL} = R_{out}^F = R_L \parallel R_C \text{ perché non viene influenzata dal feedback!}$$

$$\Rightarrow R_{out} = (1/R_{out}^F - 1/R_L)^{-1} = R_C$$

3) Considering $C_c = 5 \text{ pF}$ not negligible, find f_H , the high frequency bandwidth limit of the amplifier (internal capacitances of Q_1 and Q_2 can still be neglected).

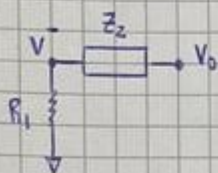
C_c e C_s non li considero perché $C_c, C_s \gg C_c$ (sono dei cortocircuiti).

$$f_H^{OL} = f_H^F = \frac{1}{2\pi} \left(\frac{1}{r_{c1}} + \frac{1}{r_{c2}} + \frac{1}{r_{c3}} \right) \approx \frac{1}{2\pi r_{c1}} = \frac{1}{2\pi C_c R_C^0} = \frac{1}{2\pi C_c R_C \parallel R_L} = 5,836 \text{ MHz}$$

E' un OpAmp invertente quindi

$$A_v^{IDEAL}(s) = \frac{V_o}{V_s} = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1} \frac{1}{1+sR_2C_2}$$

Calcolo l'espressione della rete β :

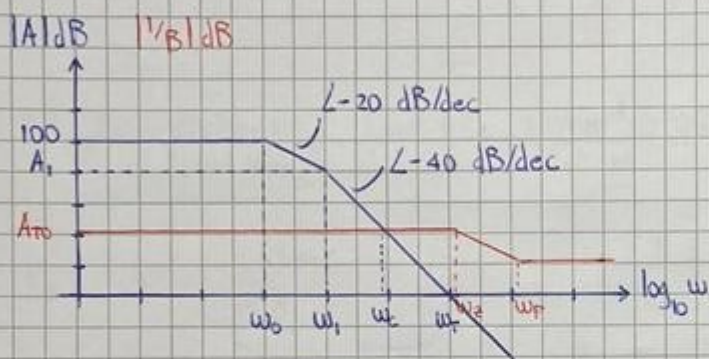


$$\beta \triangleq \frac{V_-}{V_o} \Big|_{V_s=0} = \frac{R_1}{Z_2 + R_1} = R_1 \cdot \frac{1}{R_1 + \frac{R_2}{1+sR_2C_2}} =$$

$$= \frac{R_1 + sR_1R_2C_2}{R_1 + R_2 + sR_1R_2C_2} = \frac{R_1}{R_1 + R_2} \frac{1+sR_2C_2}{1+sC_2R_1//R_2}$$

ritorno poli e
zen a β

$$A_T \triangleq \frac{1}{\beta} = \left(1 + \frac{R_2}{R_1}\right) \frac{1+sC_2R_1//R_2}{1+sR_2C_2} = A_{T0} \cdot \frac{1+s/\omega_p}{1+s/\omega_z}$$



$$A_{0dB} = 20 \log A_0 = 100 \text{ dB}$$

$$\omega_p = \frac{1}{R_1//R_2 \cdot C_2} = 2,3 \cdot 10^7 \text{ rad/s}$$

$$\omega_z = \frac{1}{R_2 C_2} = 2,1 \cdot 10^6 \text{ rad/s}$$

$$A_{T0} = 1 + \frac{R_2}{R_1} = 11 = 20,8 \text{ dB}$$

Stima approssimativa PH: opening/closing ratios = $\pm 40 \text{ dB/dec} \leadsto \text{PH} \rightarrow 0^\circ$

Per una stima accurata del PH:

$$A_1 \omega_1^2 = A_{T0} \omega_c^2 \leadsto \omega_c = \sqrt{\frac{A_1 \omega_1^2}{A_{T0}}} = 3 \cdot 10^5 \text{ rad/s}$$

$\beta \approx \omega \approx 1/\beta$

$$\text{PH} \triangleq 180^\circ + \angle T(j\omega_c) = 180^\circ + \angle A(j\omega_c) + \angle \beta(j\omega_c) =$$

$$= 180^\circ + \angle \frac{1}{(1+j\frac{\omega_c}{\omega_0})(1+j\frac{\omega_c}{\omega_1})} + \angle \frac{1+j\frac{\omega_c}{\omega_z}}{1+j\frac{\omega_c}{\omega_p}} =$$

$$= 180^\circ - \arctg\left(\frac{\omega_c}{\omega_0}\right) - \arctg\left(\frac{\omega_c}{\omega_1}\right) + \arctg\left(\frac{\omega_c}{\omega_z}\right) - \arctg\left(\frac{\omega_c}{\omega_p}\right) =$$

$$= 180^\circ - 89,8^\circ - 88,1^\circ + 8,13^\circ - 0,75^\circ = 9,5^\circ$$

2) Determine the value of capacitor C_2 for which the amplifier has $\text{PH} = 45^\circ$

Per $\text{PH} = 45^\circ$ mi serve: closing ratio = -40 dB/dec e opening ratio = $+20 \text{ dB/dec}$

Posiziono quindi lo zero della β -network in ω_c :

$$\omega_z = \frac{1}{R_2 C_2} = \omega_c \leadsto C_2 = \frac{1}{R_2 \omega_c} = 33,3 \text{ pF}$$

Più precisamente: impongo $\omega_p = \omega_T$ per avere il suo contributo alla fase trascurabile e calcolo ω_z con il GBP:

$$\pm \omega_T^2 = A_1 \cdot \omega_i^2 \quad \leadsto \quad \omega_T = \sqrt{A_1 \cdot \omega_i^2} = 1 \cdot 10^6 \text{ rad/s}$$

$$\omega_z = \frac{\omega_T}{1 + R_2/R_1} = 9,09 \cdot 10^4 \text{ rad/s}$$

$$C_2 = \frac{1}{\omega_z \cdot R_2} = 110 \text{ pF}$$

3) Using the formula below, determine the relative voltage gain error at $\omega = 0$ due to specified $A(s)$.

$$GE = \frac{A_V^{\text{IDEAL}}(0) - A_V(0)}{A_V^{\text{IDEAL}}(0)} \cdot 100$$

$$A_V^{\text{IDEAL}}(0) = -R_2/R_1 = -10$$

$$A_V^{\text{REAL}}(s) = W_{IN}(s) A_F(s)$$

$$\begin{aligned} W_{IN}(s) &= -\frac{Z_2}{Z_1 + Z_2} = -\frac{R_2}{1 + sR_2C_2} \cdot \frac{1}{R_1 + \frac{1}{1 + sR_2C_2}} = -\frac{R_2}{1 + sR_2C_2} \cdot \frac{1 + sR_2C_2}{R_1 + R_2 + sR_1R_2C_2} = \\ &= -\frac{R_2}{R_1 + R_2} \cdot \frac{1}{1 + sC_2 R_1 // R_2} \end{aligned}$$

$$A_F(s) = \frac{A_{OL}(s)}{1 + B(s)A_{OL}(s)}$$

$$A_V^{\text{REAL}}(0) = W_{IN}(0) A_F(0) = -\frac{R_2}{R_1 + R_2} \cdot \frac{A_{DO}}{1 + \frac{R_1}{R_1 + R_2} A_{DO}} = -9,999$$

$$\leadsto GE = \frac{-10 + 9,999}{-10} \cdot 100 = 0,011 \%$$